

Notes: Benmelech and Bergman (JFE, 2009)

Collateral Pricing

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1 Big picture

- **Question.** How does collateral affect the cost of debt capital? More specifically, does the liquidation value or redeployability of pledged assets lower credit spreads and expand debt capacity?
- **Empirical challenge.** A simple comparison between secured and unsecured debt is hard to interpret because riskier borrowers are often required to pledge collateral. This extensive-margin comparison can generate a misleading positive relation between collateral and loan rates.
- **Core strategy.** The paper studies the intensive margin of collateral quality within secured airline debt. All tranches are collateralized by aircraft, but the pledged aircraft differ in how easy they are to redeploy to other operators.
- **Setting.** The authors use U.S. airline secured debt, especially equipment trust certificates (ETCs), pass-through certificates (PTCs), and enhanced equipment trust certificates (EETCs), issued from 1990 to 2005.
- **Measurement contribution.** The paper links debt tranches to the exact aircraft collateral using prospectuses and aircraft serial numbers. It then constructs redeployability measures from the worldwide aircraft market.
- **Main result.** Debt backed by more redeployable aircraft has lower credit spreads, better credit ratings, and higher loan-to-value ratios.
- **Economic magnitude.** Moving from the 25th to the 75th percentile of collateral redeployability lowers credit spreads by roughly 52–64 basis points in the airline fixed-effect specifications, a large decline relative to an average spread near 199 basis points.
- **Debt capacity result.** More redeployable collateral supports higher loan-to-value ratios. The 25th–75th percentile movement in redeployability increases LTV ratios by about 9–13 percentage points.
- **Maturity result.** The paper does not find a statistically significant relation between redeployability and tranche maturity.
- **Takeaway.** Collateral is not just a binary feature of debt contracts. The quality of collateral, especially its redeployability in secondary markets, is priced by creditors and rating agencies.

2 Incremental contribution of the paper

- **From collateral use to collateral quality.** Prior empirical work often asks whether secured loans have different yields from unsecured loans. This paper shifts the focus to variation in the quality of pledged collateral among secured loans.
- **A clean industry setting.** Airlines are a useful laboratory because aircraft are large, movable, identifiable assets and secured aircraft debt is common across the industry.
- **Asset-level collateral measurement.** The authors identify the precise aircraft pledged to each tranche. This allows the analysis to connect the characteristics of collateral to the pricing and structure of debt.

- **Redeployability as a liquidation-value proxy.** The paper operationalizes the Shleifer-Vishny idea that assets are more valuable in liquidation when many potential buyers can use them.
- **Multiple debt outcomes.** The paper studies not only spreads, but also credit ratings, loan-to-value ratios, and maturity.
- **Selection analysis.** The paper directly tests whether riskier airlines pledge more redeployable collateral and finds little evidence for this concern.
- **Economic interpretation.** The results link collateral redeployability to lower expected loss given default, improved debt capacity, and lower external financing costs.

3 Data and measurement

3.1 Debt sample

- The initial sample comes from SDC Platinum and includes secured bonds, ETCs, PTCs, and EETCs issued by firms in scheduled air transportation, air courier services, and nonscheduled air transport.
- The sample period is January 1990 through December 2005.
- The authors start with 426 debt tranches. Public tranches are matched to prospectuses; private placements are supplemented using Airfinance Journal.
- The final working sample contains 246 tranches with identified collateral, issued by 12 U.S. airlines, with a total nominal value of about \$38.1 billion.
- Tranche variables include issue size, credit spread, maturity, seniority, call provisions, private-placement status, credit rating, and cumulative loan-to-value ratio.

3.2 Collateral data

- For public tranches, the paper obtains aircraft serial numbers from filing prospectuses.
- These serial numbers are matched to the Ascend CASE database, which contains worldwide commercial aircraft information.
- The sample contains 1,086 aircraft across 22 aircraft types.
- Important aircraft types in the sample include Boeing 737 (NG), Airbus A319, Boeing 757, Airbus A320, Boeing 767, and Embraer ERJ-145.

3.3 Redeployability measures

The key explanatory variables measure how easily creditors can sell or redeploy the aircraft collateral if the borrower defaults. The intuition is that collateral is more valuable when many solvent operators already use the same aircraft type.

For aircraft type s in year t , the paper constructs three market-level measures:

- Number of aircraft of the same type.
- Number of operators of the same type.
- Number of operators with at least five aircraft of the same type.

For a tranche collateral pool, the paper aggregates aircraft-type redeployability using aircraft-size weights based on seats:

$$\begin{aligned}
Redeployability_{i,t}^{aircraft} &= \sum_s \omega_{i,t,s} Redeployability_{s,t}^{aircraft}, \\
Redeployability_{i,t}^{operators} &= \sum_s \omega_{i,t,s} Redeployability_{s,t}^{operators}, \\
Redeployability_{i,t}^{operators>5} &= \sum_s \omega_{i,t,s} Redeployability_{s,t}^{operators>5}.
\end{aligned}$$

- Bankrupt airlines are excluded from the set of potential buyers.
- The issuing airline's own aircraft or own operator status is subtracted when calculating residual secondary-market demand.
- The three measures are highly intuitive proxies for expected collateral value in default.

3.4 Airline controls

- Airline size is measured as the log of book assets.
- Market-to-book captures valuation and growth opportunities.
- Profitability is operating income divided by assets.
- Leverage and interest coverage are used in the selection tests.
- Airline credit rating controls for borrower risk.

4 Models and empirical specifications

4.1 Theoretical predictions

The paper develops four predictions about collateral liquidation values:

- **Prediction 1.** Credit spreads decrease in collateral liquidation values, holding other factors constant.
- **Prediction 2.** Tranche credit ratings improve in collateral liquidation values.
- **Prediction 3.** Loan-to-value ratios increase in collateral liquidation values.
- **Prediction 4.** Tranche maturity increases in collateral liquidation values.

The first three predictions receive strong empirical support; the fourth does not.

4.2 Credit-spread regressions

The baseline spread specification is:

$$Spread_{i,a,t} = \beta Redeployability_{i,a,t} + X'_{i,a,t}\gamma + Z'_{a,t-1}\delta + c_t + \varepsilon_{i,a,t},$$

where i indexes the tranche, a the airline, and t the year.

- $Spread_{i,a,t}$ is the credit spread over the corresponding Treasury yield.
- $X_{i,a,t}$ includes tranche controls: seniority, tranche size, call provision, private placement, and maturity.
- $Z_{a,t-1}$ includes airline controls: size, market-to-book, profitability, and airline credit rating.
- c_t are year fixed effects.

The richer specification adds airline fixed effects:

$$Spread_{i,a,t} = \beta Redeployability_{i,a,t} + X'_{i,a,t}\gamma + Z'_{a,t-1}\delta + c_t + b_a + u_{i,a,t}.$$

4.3 Industry downturn interactions

The paper tests whether redeployability matters more when the airline industry is weak. The logic is from Shleifer and Vishny: when an industry is distressed, potential buyers are financially constrained, so redeployability should be especially valuable.

The interaction specification relates spreads to redeployability, industry profitability, and their interaction. The expected sign is that the spread-reducing effect of redeployability is stronger when industry profitability is low.

4.4 Credit-rating regressions

Credit ratings are ordinal, so the paper uses ordered probit models. The dependent variables are Moody's and S&P tranche ratings, numerically ordered so that better ratings have lower numeric values.

- A negative coefficient on redeployability means higher redeployability improves ratings.
- The paper reports coefficients rather than marginal effects in Table 9.
- Table 11 translates these estimates into economically interpretable changes in rating probabilities.

4.5 Loan-to-value and maturity regressions

For loan-to-value, the paper estimates both tranche-level and issue-level specifications:

$$LTV_{i,a,t}^T = \beta Redeployability_{i,a,t} + X'_{i,a,t}\gamma + Z'_{a,t-1}\delta + c_t + b_a + u_{i,a,t},$$

$$LTV_{j,a,t}^I = \beta Redeployability_{j,a,t} + Z'_{a,t-1}\delta + c_t + b_a + u_{j,a,t}.$$

For maturity, the paper estimates:

$$Maturity_{i,a,t} = \beta Redeployability_{i,a,t} + X'_{i,a,t}\gamma + Z'_{a,t-1}\delta + c_t + b_a + u_{i,a,t}.$$

4.6 Robustness controlling for loan-to-value

Because higher LTV mechanically raises creditor risk and can affect spreads, the paper also estimates spread regressions that include cumulative loan-to-value:

$$Spread_{i,a,t} = \beta Redeployability_{i,a,t} + \lambda LTV_{i,a,t} + X'_{i,a,t}\gamma + Z'_{a,t-1}\delta + c_t + b_a + \xi_{i,a,t}.$$

The redeployability coefficients remain negative and significant after controlling for LTV.

5 Identification and empirical design

5.1 Why the intensive margin helps

- A standard secured-versus-unsecured comparison is confounded because weak borrowers are more likely to pledge collateral.
- This paper avoids much of that problem by comparing collateral quality among secured debt instruments.
- Since all tranches in the core sample are secured by aircraft, the key variation is whether the aircraft are easier or harder to redeploy.

5.2 Why airlines and aircraft are useful

- Aircraft are movable, standardized, and identifiable assets.
- The secondary market for aircraft is observable through worldwide operator data.
- ETCs and EETCs are asset-backed instruments in which the collateral is central to creditor recovery.
- Section 1110 of the U.S. Bankruptcy Code gives secured aircraft creditors relatively strong repossession rights, making collateral value especially relevant.

5.3 Selection concerns

The main remaining concern is intensive-margin selection: riskier airlines might be forced to pledge more redeployable aircraft. This would bias the analysis against finding the predicted negative relation between redeployability and spreads, because redeployability would proxy for higher borrower risk.

The paper addresses this concern by regressing redeployability on airline risk variables such as size, profitability, leverage, interest coverage, and S&P airline credit rating. Table A1 shows little evidence that riskier airlines pledge more redeployable collateral.

5.4 What the design can identify

- The estimates identify how the market prices observable variation in the redeployability of collateral within airline secured debt.

- The design is not a randomized experiment, but the intensive-margin setting, detailed collateral measurement, and fixed effects make the interpretation substantially cleaner than secured-versus-unsecured comparisons.
- The results are best read as strong evidence that creditors and rating agencies price asset redeployability at issuance.

6 Tables and figures

6.1 Table 1: Selected EETC transactions

Read:

- Table 1 describes three EETC issues by FedEx, Northwest Airlines, and Delta.
- Each issue has multiple tranches backed by the same aircraft pool but differing in seniority, spread, rating, maturity, and cumulative LTV.
- Senior tranches receive better ratings and lower spreads than junior tranches, even when they share the same underlying collateral.
- The table introduces the institutional structure of EETCs: tranche seniority separates claims on the same aircraft pool.

Economic message: The table shows why tranche-level controls are essential. Collateral is common within an issue, but seniority and LTV strongly affect pricing and ratings.

6.2 Table 2: Equipment trust certificates issuance

Read:

- Table 2 summarizes ETC, PTC, and EETC issuance by airline and period.
- The sample contains 246 tranches with total nominal value of \$38.1 billion.
- Issuance is largest in 1998–2001, with 108 tranches and \$23.7 billion.
- Continental, American, Delta, United, FedEx, and U.S. Airways are major issuers.

Economic message: Secured aircraft debt is economically important for airline financing. The paper studies a major source of external capital rather than a niche contract.

Table 1

Selected EETC transactions.

This table displays the characteristics of three EETC issues by FedEx, Northwest Airlines, and Delta Airlines. Tranche is the name of each individual tranche (presented by seniority). Issue size is the dollar value (in \$ millions) of the tranche. Yield is the yield of the tranche at the time of issue. Spread is credit spread (in basis points) over its corresponding treasury yield. Maturity is the number of years until the final payment. Moody's tranche rating and S&P tranche rating are the ratings of the tranche assigned by either Moody's or S&P. LTV is the tranche cumulative loan-to-value ratio defined as the ratio between the sum of the principal amount of that tranche and all tranches senior to it, divided by an appraisal of the value of the assets serving as collateral. Collateral provides a description of the aircraft serving as tranche collateral.

EETC	Tranche	Issue size	Yield (%)	Spread (bp)	Maturity	Moody's rating	S&P rating	LTV	Collateral
FedEx 1998-1	1-A	458.1	6.720	125	1/2022	Aa2	AAA	0.387	5 MD-11F 8 A300F4-605R
FedEx 1998-1	1-B	178.6	6.845	138	1/2019	A1	AA-	0.532	5 MD-11F 8 A300F4-605R
FedEx 1998-1	1-C	196.8	7.020	155	1/2016	Baa1	BBB+	0.688	5 MD-11F 8 A300F4-605R
NWA 1999-3	G	150.2	7.935	170	6/2019	Aaa	AAA	0.441	14 BAE Avro RJ85
NWA 1999-3	B	58.6	9.485	325	6/2015	Baa2	BBB	0.614	14 BAE Avro RJ85
NWA 1999-3	C	30.5	9.152	300	6/2010	Baa3	BBB-	0.691	14 BAE Avro RJ85
Delta 2002-1	G-1	586.9	6.718	153	1/2023	Aaa	AAA	0.519	17 B737-832 1 B757-232 8 B767-332ER 6 B767-432ER
Delta 2002-1	G-2	370.3	6.417	123	7/2012	Aaa	AAA	0.519	17 B737-832 1 B757-232 8 B767-332ER 6 B767-432ER
Delta 2002-1	C	168.7	7.779	325	1/2012	Baa2	A-	0.611	17 B737-832 1 B757-232 8 B767-332ER 6 B767-432ER

Table 1: Selected EETC transactions.

Table 2

Equipment trust certificates issuance.

This table displays the distribution of nominal values of 246 equipment trust certificates (ETCs), pass-through certificates (PTCs), and enhanced equipment trust certificates (EETCs) that were issued in the U.S. public markets between 1990 and 2005 and are included in our sample. Number of issues per airline are provided in parentheses for different sample periods.

Airline	1990-1993	1994-1997	1998-2001	2002-2005	1990-2005
Alaska Airlines	\$98.9 (4)				\$98.9 (4)
America West		\$93.8 (4)	\$1,079.9 (9)		\$1,173.7 (13)
American Airlines	\$668.9 (23)	\$65.9 (1)	\$3,792.4 (14)	\$871.9 (2)	\$5,398.1 (40)
Atlas Air			\$543.5 (4)		\$543.5 (4)
Continental Airlines		\$2,287.7 (19)	\$5,338.0 (29)	\$1,575.5 (7)	\$9,201.2 (55)
Delta Air Lines	\$1,305.5 (10)	\$79.6 (2)	\$2,748 (8)	\$1,125.9 (3)	\$5,059.0 (23)
Federal Express	\$41.9 (6)	\$1,615.7 (14)	\$1,183.6 (7)		\$3,341.2 (27)
JetBlue Airways				\$929.3 (6)	\$929.3 (6)
Southwest Airlines	\$168.0 (3)	\$610.4 (9)	614.3 (3)		\$1,392.7 (15)
United Air Lines	\$724.3 (12)	\$610.6 (3)	\$3,622.5 (12)		\$4,957.4 (27)
U.S. Airways	\$168.6 (3)	\$263.0 (3)	\$2,947.3 (10)		\$3,378.9 (16)
Northwest Airlines			\$1,872.8 (12)	\$749.1 (4)	\$2,621.9 (16)
Total	\$3,476.1 (61)	\$5,626.7 (56)	\$23,742.3 (108)	\$5,251.7 (22)	\$38,096.8 (246)

Table 2: Equipment trust certificates issuance.

6.3 Table 3: Summary statistics

Read:

- Table 3 reports tranche and airline characteristics.
- The average tranche size is \$154.9 million; the mean spread is 198.8 basis points.
- The average cumulative loan-to-value ratio is 0.616, and the average maturity is 15.4 years.
- The average issuing airline has about \$10 billion in assets, profitability of 11.7 percent, and leverage of 0.39.
- Tranche ratings are generally much better than airline ratings, consistent with collateral improving credit quality.

Economic message: The data contain meaningful variation in spreads, ratings, LTV, maturity, and airline risk, allowing the paper to separate collateral effects from borrower and contract features.

6.4 Table 4: Aircraft type

Read:

- Table 4 lists the 22 aircraft types used as collateral.
- The most common collateral type is Boeing 737 (NG), followed by Airbus A319, Boeing 757, Airbus A320, Boeing 767, and Embraer ERJ-145.
- The table reports the time-series average of the redeployability measures for each aircraft type.
- Boeing 737 (CFMI) and Boeing 737 (NG) have many operators, while niche regional aircraft have far fewer potential buyers.

Economic message: Redeployability is not abstract. It reflects concrete aircraft-market depth: how many aircraft exist and how many airlines can plausibly absorb a repossessed plane.

Table 3**Summary statistics.**

This table provides descriptive statistics for the variables used in the empirical analysis. Panel A displays tranche characteristics. Panel B provides airlines characteristics. Tranche size is the dollar value (in \$ millions) of the tranche. Number of aircraft is the number of aircraft serving as collateral. Spread is credit spread (in basis points) over its corresponding treasury yield. Loan-to-value is the ratio between the cumulative book value of the loan and the appraised value of the aircraft used as collateral for the loan. S&P tranche rating and Moody's tranche rating are the ratings of the tranche assigned by either S&P or Moody's. Maturity is the number of years until the final payment. Seniority is the tranche seniority (1 = most senior). Call provision is a dummy variable that equals to one if the tranche is callable. Private is a dummy variable that equals to one for private placement tranches. Size is the book value of the airline assets. Market-to-book is calculated as the market value of equity minus the book value of equity, all over the book value of assets. Profitability is defined as operating income over assets. Leverage is defined as total debt divided by total assets. S&P airline credit rating is the airline long-term credit rating.

	Mean	25th Percentile	Median	75th Percentile	Standard deviation	Min	Max
Panel A: Tranche characteristics							
Tranche size (\$m)	154.87	47.6	112.5	194.5	148.5	0.7	1,319.6
Number of aircraft	16.2	4.0	13.0	21.0	11.9	1.0	46.0
Spread	198.8	135.0	182.0	238.0	107.5	40.0	909.4
Loan-to-value	0.616	0.490	0.600	0.780	0.329	0.000	0.800
S&P tranche rating	N/A	A	BBB+	AA+	N/A	B+	AAA
Moody's tranche rating	N/A	A3	Baa2	Aa3	N/A	B1	Aaa
Maturity	15.4	10.3	16.3	20.6	6.1	1.7	25.0
Seniority	1.50	1.00	1.00	2.00	0.80	1.00	4.00
Call provision	0.14	0.00	0.00	0.00	0.35	0.00	1.00
Private	0.17	0.00	0.00	0.00	0.38	0.00	1.00
Panel B: Airlines characteristics							
Size (\$m)	10,024.4	5,793.1	8,768.8	10,877.4	6,800.9	1,211.6	32,841.0
Market-to-book	1.19	1.02	1.15	1.22	0.27	0.86	2.51
Profitability (%)	11.7	8.97	12.68	14.26	7.21	-7.26	46.17
Leverage	0.39	0.32	0.39	0.44	0.11	0.13	0.67
S&P airline credit rating	N/A	BB+	BB-	BBB	N/A	B-	A

Table 3: Summary statistics.

Table 4**Aircraft type.**

This table lists the 22 different aircraft types represented in our sample. For every aircraft type we list the number of aircraft and percentage of total aircraft in our sample. The table also reports the time-series (1990-2005) mean of the three redeployability measures for every aircraft type. Redeployability (# of aircraft) is the number of aircraft per type; Redeployability (# of operators) is the number of operators per type; Redeployability (# of operators with more than 5 aircraft) is the number of operators who operate at least five aircraft per type.

Aircraft type	Number	Percent	(time-series mean) # of aircraft	(time-series mean) # of operators	(time-series mean) # operators with > 5 aircraft
Airbus A300	25	2.30	278.9	53.1	17.8
Airbus A310	12	1.10	161.7	44.9	10.6
Airbus A319	144	13.26	188.7	36.3	8.9
Airbus A320	89	8.20	621.1	83.2	34.1
Airbus A321	10	0.92	116.6	31.0	9.1
Airbus A330	9	0.83	117.5	30.7	9.1
BAE SYSTEMS RJ Avroliner	14	1.29	44.1	6.0	2.1
BAE SYSTEMS Jetstream 31/53	2	0.18	180.9	64.0	8.5
Boeing (McDonnell-Douglas) MD-11	24	2.21	130.4	26.6	9.8
Boeing (McDonnell-Douglas) MD-80	48	4.51	997.7	67.3	28.3
Boeing 737 (CFM)	58	5.34	1502.3	162.9	60.0
Boeing 737 (NG)	222	20.44	686.1	143.3	28.7
Boeing 747	21	1.93	760.8	91.4	36.3
Boeing 757	141	12.98	688.0	80.0	26.2
Boeing 767	80	7.37	617.6	83.5	30.1
Boeing 777	56	5.16	185.8	25.8	9.9
Bombardier Dash 8	2	0.18	389.4	69.0	18.9
Embraer EMB-120	9	0.83	31.2	5.0	1.3
Embraer E175	8	0.74	19.3	6.7	0.7
Embraer E175	98	9.02	205.7	20.1	7.2
Lockheed L-1011 TriStar	11	1.01	73.8	28.5	4.4
Saab 340	2	0.18	274.0	40.1	12.5
Total	1,086	100.00	-	-	-

Table 4: Aircraft type.

6.5 Table 5: Redeployability measures

Read:

- Panel A summarizes the three redeployability measures at the tranche-collateral level.
- On average, collateral aircraft types have 958.6 aircraft, 118.7 operators, and 40.2 operators with more than five aircraft.
- Panel B gives examples: Northwest's BAE RJ85 collateral is much less redeployable than Southwest's Boeing 737-300 collateral.
- The examples make the source of identifying variation intuitive: collateral pools differ sharply in the depth of the secondary market.

Economic message: Creditors should value collateral differently when there are many plausible secondary-market buyers. Table 5 translates that idea into measurable variables.

6.6 Table 6: Collateral value and credit spread

Read:

- Table 6 is the core spread result.
- In all specifications, redeployability is negatively related to credit spreads.
- With year and airline fixed effects, the coefficients become larger in absolute value and remain statistically significant.
- A 25th-to-75th percentile increase in redeployability lowers spreads by roughly 52–64 basis points.
- Seniority and private-placement status also matter: lower seniority and private placements are associated with higher spreads.

Economic message: More redeployable collateral reduces creditors' expected loss given default and is capitalized into lower promised yields at issuance.

Table 5
Redeployability measures.

This table provides descriptive statistics for the redeployability measures used in the empirical analysis. Panel A displays the characteristics of the redeployability measures. Redeployability (# of aircraft) is the number of aircraft per type. Redeployability (# of operators) is the number of operators per type. Redeployability (# of operators with more than 3 aircraft) is the number of operators who operate at least five aircraft per type. Panel B presents example of specific tranches, the collateral used to secure the tranches, and the redeployability measures values for each of the tranches.

	Mean	25th Percentile	Median	75th Percentile	Standard deviation	Min	Max
Panel A: Summary statistics							
Redeployability (# of aircraft)	958.6	365.0	951.7	1624.1	793.8	70.0	3,456.0
Redeployability (# of operators)	108.7	53.2	73.6	165.8	90.7	7.0	421.0
Redeployability (# of operators with > 3 aircraft)	48.2	17.6	27.0	63.0	30.0	2.0	133.0
Panel B: Example of tranche redeployability							
ETEC	Tranche	Issue size	Spread (bp)	# of aircraft	# of operators	# of operators with > 3 aircraft	Collateral
Delta 2000-1	A1	340.1	148	1657.2	185.8	63.0	20 8757-932
							18 8751-222
							6 8757-3228
							18 8757-232
Delta 2000-1	A2	738.1	170	1657.2	185.8	63.0	20 8757-932
							18 8751-222
							6 8757-3228
							18 8757-232
Delta 2000-1	B	182.5	205	1657.2	185.8	63.0	20 8757-932
							18 8751-222
							6 8757-3228
							18 8757-232
Delta 2000-1	C	238.3	188	1657.2	185.8	63.0	20 8757-932
							18 8751-222
							6 8757-3228
							18 8757-232
FWA 1999-3	C	150.2	170	72.0	7.0	1.0	14 BAA 8805
							14 BAA 8805
							14 BAA 8805
							14 BAA 8805
FWA 1999-3	B	361.0	325	72.0	7.0	1.0	14 BAA 8805
							14 BAA 8805
							14 BAA 8805
							14 BAA 8805
FWA 1999-3	C	33.9	300	72.0	7.0	1.0	14 BAA 8805
							14 BAA 8805
							14 BAA 8805
							14 BAA 8805
Southwest 1998	A1	115.1	72	2046.0	380.0	112.0	6 8757-300
							6 8757-300
							6 8757-300
							6 8757-300
Southwest 1998	A2	33.1	85	2046.0	380.0	112.0	6 8757-300
							6 8757-300
							6 8757-300
							6 8757-300

Table 5: Redeployability measures.

Table 6
Collateral value and credit spread.

The dependent variable in the regressions is credit spread (in basis points) over its corresponding treasury yield. Seniority is the tranche seniority (1 = most senior, Private is a dummy variable that equals one if the tranche is callable. Maturity is the number of years until the final payment. Airline size is the logarithm of the book value of the airline assets. Market-to-book is calculated as the market value of equity minus the book value of equity, all over the book value of assets. Profitability is defined as operating income over assets. S&P Airline credit rating is the airline long-term credit rating. Redeployability (aircraft) is the number of aircraft per type; Redeployability (operators) is the number of operators per type. Redeployability (> 3 aircraft) is the number of operators who operate at least five aircraft per type. All regressions include an intercept (not reported) and year fixed effects. t -statistics are calculated using standard-errors that are clustered by airline and reported in parentheses.

Dependent variable =	Credit spread	Credit spread	Credit spread	Credit spread	Credit spread	Credit spread
Seniority	54.17 (5.67)	54.34 (5.71)	54.11 (5.68)	54.49 (5.44)	54.26 (5.36)	54.19 (5.36)
Private	37.13 (3.65)	34.61 (3.47)	34.74 (3.37)	32.36 (2.87)	25.39 (2.07)	27.80 (2.42)
Tranche size	1.84 (0.22)	2.41 (0.29)	2.25 (0.27)	4.03 (0.52)	4.49 (0.55)	4.12 (0.52)
Call provision	72.61 (3.28)	71.73 (3.17)	71.75 (3.15)	68.77 (3.62)	70.01 (3.39)	68.61 (3.32)
Maturity	0.35 (0.20)	0.32 (0.18)	0.27 (0.15)	0.49 (0.26)	0.49 (0.26)	0.51 (0.27)
Airline size	-14.95 (-2.66)	-15.67 (-2.70)	-15.87 (-2.76)	93.75 (2.11)	88.05 (1.91)	85.88 (1.97)
Market-to-book	6.94 (0.24)	8.59 (0.30)	8.93 (0.32)	71.36 (1.90)	76.61 (2.07)	67.34 (1.90)
Profitability	-259.55 (-2.33)	-264.07 (-2.36)	-259.51 (-2.30)	-459.29 (-3.60)	-487.05 (-3.60)	-492.59 (-3.67)
S&P airline credit rating	5.69 (1.77)	5.68 (1.78)	5.58 (1.74)	2.18 (0.39)	1.79 (0.39)	1.94 (0.45)
Redeployability (aircraft)	-0.018 (-2.24)			-0.051 (-2.85)		
Redeployability (operators)		-0.163 (-2.33)			-0.394 (-2.41)	
Redeployability (> 3 aircraft)			-0.523 (-2.39)			-1.277 (-3.18)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Airline fixed effects	No	No	No	Yes	Yes	Yes
Adjusted R^2	0.60	0.60	0.60	0.66	0.66	0.66
Observations	225	225	225	225	225	225

Table 6: Collateral value and credit spread.

6.7 Figure 1: The cyclicality of the airline industry

Read:

- Figure 1 plots industry profitability and market-to-book ratios from 1990 to 2005.
- Industry conditions improved through the mid-1990s and peaked around 1997.
- Profitability collapsed sharply around 2001–2002 and partially recovered afterward.
- The figure motivates the downturn tests: collateral redeployability may be especially valuable when the whole industry is distressed.

Economic message: Industry-wide distress makes liquidation harder because potential buyers are also financially constrained. This is precisely when redeployability should matter most.

6.8 Table 7: Industry conditions and credit spread

Read:

- Table 7 replaces year fixed effects with industry condition controls.
- Better industry conditions are associated with lower spreads.
- The negative relation between redeployability and spreads remains after controlling for industry profitability or industry market-to-book.
- The coefficients are larger in absolute value than in the year fixed-effect specifications, suggesting that time variation captures important industry-cycle effects.

Economic message: Collateral redeployability is not just a proxy for the airline cycle. It continues to predict spreads when industry conditions are explicitly controlled.

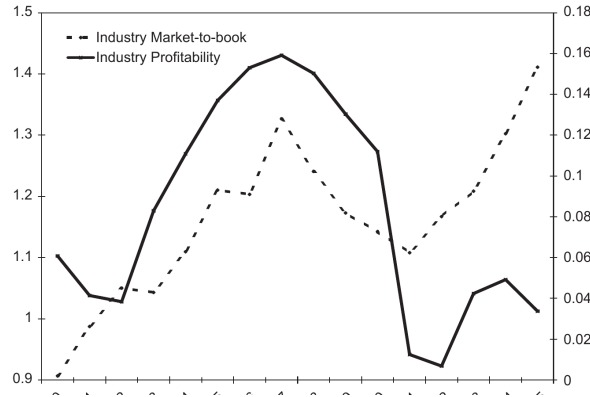


Figure 1: The cyclical nature of the airline industry.

Table 7
Industry conditions and credit spread.
The dependent variable in the regressions is credit spread (in basis points) over its corresponding treasury yield. Seniority is the tranche seniority (1 = most senior). Tranche size is the logarithm of the dollar value (in \$ millions) of the tranche. Private is a dummy variable that equals one for private placement tranches. Call provision is a dummy variable that equals one if the tranche is callable. Maturity is the number of years until the final payment. Airline size is the logarithm of the book value of the airline assets. Market-to-book is calculated as the market value of equity minus the book value of equity, all over the book value of assets. Profitability is defined as operating income over assets. S&P airline credit rating is the airline long-term credit rating. Industry profitability is a weighted average profitability at the airline industry level. Industry market-to-book is a weighted average industry market-to-book. Redeployability (aircraft) is the number of aircraft per type; Redeployability (operators) is the number of operators per type. Redeployability (> 5 aircraft) is the number of operators who operate at least five aircraft per type. All regressions include an intercept (not reported) and airline fixed effects. t-statistics are calculated using standard-errors that are clustered by airline and reported in parentheses.

Dependent variable =	Credit spread	Credit spread	Credit spread	Credit spread	Credit spread	Credit spread
Seniority	42.92 (4.61)	42.99 (4.44)	42.85 (4.43)	40.50 (4.23)	40.39 (4.06)	40.77 (4.14)
Private	18.38 (0.51)	2.85 (0.07)	9.76 (0.28)	23.86 (0.62)	6.88 (0.16)	13.31 (0.36)
Tranche size	-5.75 (-1.09)	-5.29 (-1.02)	-6.02 (-1.17)	-6.18 (-1.04)	-5.75 (-1.01)	-6.38 (-1.14)
Call provision	22.53 (0.90)	23.42 (0.92)	24.10 (0.97)	28.51 (1.43)	29.94 (1.40)	29.47 (1.40)
Maturity	1.14 (0.61)	1.16 (0.63)	1.12 (0.61)	1.00 (0.55)	1.03 (0.57)	0.99 (0.54)
Airline size	125.31 (2.51)	121.84 (2.55)	117.56 (2.44)	145.66 (2.66)	141.34 (2.62)	134.56 (2.57)
Market-to-book	46.98 (1.82)	56.82 (2.31)	36.22 (1.35)	55.62 (1.83)	67.45 (2.29)	43.40 (1.34)
Profitability	115.46 (0.84)	81.59 (0.60)	60.75 (0.50)	-201.87 (-1.43)	-252.18 (-2.01)	-219.03 (-1.87)
S&P airline credit rating	-4.73 (-1.09)	-5.17 (-1.20)	-3.70 (-0.91)	0.18 (0.04)	-0.31 (-0.06)	0.65 (0.14)
Industry profitability	-812.92 (-5.09)	-827.86 (-5.35)	-715.35 (-4.67)			
Industry market-to-book				-253.11 (-2.97)	-253.82 (-3.10)	-217.34 (-2.81)
Redeployability (aircraft)	-0.087 (-5.71)			-0.097 (-5.57)		
Redeployability (operators)		-0.716 (-5.43)			-0.793 (-5.05)	
Redeployability (> 5 aircraft)			-2.361 (-5.97)			-2.626 (-5.52)
Airline fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.47	0.47	0.48	0.45	0.45	0.46
Observations	225	225	225	225	225	225

Table 7: Industry conditions and credit spread.

6.9 Table 8: Collateral value, industry conditions, and credit spread

Read:

- Table 8 interacts redeployability with industry profitability.
- The direct redeployability effect is negative.
- The interaction with industry profitability is positive in the specifications without year fixed effects.
- This implies that the spread-reducing effect of redeployability is stronger when industry profitability is low.
- With both airline and year fixed effects, the interaction signs remain positive but lose statistical significance.

Economic message: The evidence is consistent with the liquidation-value channel being more important in downturns, although the interaction evidence is weaker when time effects are fully absorbed.

6.10 Table 9: Collateral value and credit rating

Read:

- Table 9 estimates ordered probit models for Moody's and S&P tranche ratings.
- Redeployability coefficients are negative, meaning that more redeployable collateral improves ratings because better ratings are coded with lower numeric values.
- The results are statistically significant or close to significant across the three redeployability measures.
- Seniority is strongly related to ratings, as expected.

Economic message: Rating agencies appear to incorporate collateral redeployability into their assessment of debt quality, not only borrower-level risk.

Table 8
Collateral value, industry conditions, and credit spread.

The dependent variable in the regressions is credit spread (in basis points) over its corresponding treasury yield. Industry profitability is a weighted average profitability at the airline industry level. Redeployability (aircraft) is the number of aircraft per type; Redeployability (operators) is the number of operators per type; Redeployability (>5 aircraft) is the number of operators who operate at least five aircraft per type. Regressions also include tranche characteristics (seniority, size, a dummy for private placement, and controls for call provisions at tranche maturity), and airline characteristics (airline size, market-to-book, profitability, and S&P airline long-term credit rating) which are not reported for brevity. All regressions include an intercept (not reported) and airline fixed effects. t-statistics are calculated using standard-errors that are clustered by airline and reported in parentheses.

Dependent variable =	Credit spread	Credit spread	Credit spread	Credit spread	Credit spread	Credit spread
Redeployability (aircraft)	-0.142 (-5.34)			-0.084 (-3.31)		
Redeployability (operators)		-1.208 (-5.45)			-0.668 (-3.02)	
Redeployability (>5 aircraft)			-3.347 (-4.79)			-1.908 (-3.30)
Redeployability (aircraft) × Industry profitability _{t-1}	0.461 (2.18)			0.285 (1.55)		
Redeployability (operators) × Industry profitability _{t-1}		4.27 (2.85)			2.368 (1.62)	
Redeployability (>5 aircraft) × Industry profitability _{t-1}			8.78 (1.67)			5.739 (1.25)
Tranche controls	Yes	Yes	Yes	Yes	Yes	Yes
Airline controls	Yes	Yes	Yes	Yes	Yes	Yes
Airline fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	No	No	No	Yes	Yes	Yes
Adjusted R ²	0.48	0.48	0.48	0.60	0.60	0.60
Observations	225	225	225	225	225	225

Table 8: Collateral value, industry conditions, and credit spread.

Table 9
Collateral value and credit rating.

The dependent variable in the regressions is either Moody's tranche rating or S&P tranche rating. Seniority is the tranche seniority (1 = most senior). Private is a dummy variable that equals one if the tranche is private placement tranches. Tranche size is the logarithm of the dollar value (in \$ millions) of the tranche. Call provision is a dummy variable that equals one if the tranche is callable. Maturity is the number of years until the final payment. Airline size is the logarithm of the book value of the airline assets. Market-to-book is calculated as the market value of equity minus the book value of equity, all over the book value of assets. Profitability is defined as operating income over assets. S&P airline credit rating is the airline long-term credit rating. Redeployability (aircraft) is the number of aircraft per type; Redeployability (operators) is the number of operators per type; Redeployability (>5 aircraft) is the number of operators who operate at least five aircraft per type. The table reports coefficients (not marginal effects) from ordered probit regressions that include an intercept (not reported) and year fixed effects. t-statistics are calculated using standard-errors that are clustered by airline and reported in parentheses.

Dependent variable =	Moody's credit rating	Moody's credit rating	Moody's credit rating	S&P credit rating	S&P credit rating	S&P credit rating
Ordered probit regressions						
Seniority	1.49 (7.47)	1.49 (7.57)	1.50 (7.62)	1.77 (5.71)	1.78 (5.78)	1.78 (5.82)
Private	0.18 (0.63)	0.14 (0.53)	0.14 (0.52)	-0.53 (-1.54)	-0.59 (-1.71)	-0.59 (-1.68)
Tranche size	-0.17 (-1.49)	-0.16 (-1.38)	-0.16 (-1.42)	-0.31 (-2.75)	-0.30 (-2.66)	-0.30 (-2.70)
Call provision	0.95 (2.33)	0.94 (2.27)	0.93 (2.23)	0.82 (3.00)	0.80 (2.92)	0.80 (2.87)
Maturity	-0.01 (-0.39)	-0.02 (-0.42)	-0.01 (-0.48)	-0.001 (-0.04)	-0.002 (-0.07)	-0.003 (-0.14)
Airline size	-0.49 (-2.67)	-0.50 (-2.74)	-0.51 (-2.77)	-0.37 (-2.78)	-0.38 (-2.79)	-0.39 (-2.91)
Market-to-book	-1.02 (-1.83)	-0.97 (-1.68)	-0.94 (-1.65)	-1.36 (-2.05)	-1.33 (-1.98)	-1.29 (-1.90)
Profitability	-0.14 (-0.07)	-0.20 (-0.10)	-0.10 (-0.05)	-3.44 (-1.55)	-3.50 (-1.60)	-3.44 (-1.56)
Redeployability (aircraft)	-0.0003 (-1.75)			-0.0004 (-2.32)		
Redeployability (operators)		-0.003 (-1.89)			-0.003 (-2.40)	
Redeployability (>5 aircraft)			-0.010 (-2.02)			-0.011 (-2.49)
Fixed effects	Year	Year	Year	Year	Year	Year
Pseudo R ²	0.24	0.25	0.25	0.32	0.32	0.32
Observations	227	227	227	223	223	223

Table 9: Collateral value and credit rating.

6.11 Table 10: Collateral value, loan-to-value, and maturity

Read:

- Columns 1–3 show tranche-level LTV regressions.
- Columns 4–6 show issue-level LTV regressions.
- Columns 7–9 show maturity regressions.
- Redeployability is positively related to LTV at both the tranche and issue level.
- Redeployability is not significantly related to maturity.

Economic message: More redeployable collateral supports more borrowing against the same asset pool. The paper finds a strong debt-capacity channel, but not a maturity channel.

6.12 Table 11: Economic significance of collateral redeployability

Read:

- Panel A translates spread coefficients into basis-point effects.
- A 25th–75th percentile increase in redeployability reduces spreads by 52–64 basis points.
- Panel B translates rating regressions into marginal effects on rating probabilities.
- Panel C translates LTV regressions into percentage-point effects on leverage against collateral.
- A 25th–75th percentile increase in redeployability raises LTV by about 0.09–0.13.

Economic message: The estimated effects are economically large. Redeployability materially lowers debt costs and increases debt capacity.

Table 10

Collateral value, loan-to-value, and tranche maturity.

The dependent variable in columns 1-3 is the cumulative loan-to-value (LTV) at the tranche level. The dependent variable in columns 4-6 is the cumulative loan-to-value (LTV) at the issue level. The dependent variable in columns 7-9 is the tranche maturity in years. Seniority is the tranche seniority (1 = most seniors. Private is a dummy variable that equals one if private placement tranches. Tranche size is the logarithm of the dollar value (in \$m) of the tranche. Call provision is a dummy variable that equals one if the tranche is callable. Maturity is the number of years until the final payment. Airline size is the logarithm of the book value of the airline assets. Market-to-book is calculated as the market value of equity minus the book value of equity, all over the book value of assets. Profitability is defined as operating income over assets. S&P airline credit rating is the airline long-term credit rating. Redeployability (aircraft) is the number of aircraft per type. Redeployability (operators) is the number of operators per type. Redeployability (>5 aircraft) is the number of operators who operate at least five aircraft per type. All regressions include an intercept (not reported) and airline and year fixed effects. t-statistics are calculated using standard-errors that are clustered by airline and reported in parentheses.

Model dependent variable =	(1) LTV	(2) LTV	(3) LTV	(4) LTV	(5) LTV	(6) LTV	(7) Maturity	(8) Maturity	(9) Maturity
Seniority	0.10 (12.75)	0.10 (13.26)	0.10 (12.63)	-	-	-	-1.63 (-2.91)	-1.62 (-2.83)	-1.63 (-2.90)
Private	-0.05 (-1.23)	-0.04 (-1.11)	-0.05 (-1.18)	-	-	-	0.51 (0.29)	0.32 (0.19)	0.41 (0.24)
Tranche size	0.003 (0.53)	0.002 (0.81)	0.002 (0.40)	-	-	-	2.15 (2.41)	2.17 (2.44)	2.16 (2.42)
Call provision	0.02 (0.78)	0.02 (0.81)	0.02 (0.73)	-	-	-	-0.82 (-0.68)	-0.82 (-0.66)	-0.81 (-0.67)
Maturity	-0.002 (-0.29)	-0.002 (-0.20)	-0.002 (-0.19)	-	-	-	-	-	-
Airline size	-0.21 (-1.44)	-0.22 (-1.51)	-0.19 (-1.30)	-0.004 (-0.04)	-0.01 (-0.10)	0.02 (0.20)	-4.36 (-0.82)	-4.30 (-0.80)	-4.50 (-0.82)
Market-to-book	-0.04 (-0.78)	-0.05 (-1.06)	-0.04 (-0.63)	-0.08 (-2.34)	-0.10 (-2.23)	-0.07 (-2.15)	3.11 (0.76)	3.29 (0.81)	3.04 (0.73)
Profitability	0.23 (-0.69)	-0.29 (-0.81)	-0.19 (-0.55)	-0.15 (-0.86)	-0.13 (-0.55)	-0.10 (-0.55)	-18.98 (-1.20)	-19.53 (-1.13)	-19.53 (-1.17)
S&P airline credit rating	0.01 (0.56)	0.01 (0.47)	0.01 (0.55)	-0.02 (-1.94)	-0.02 (-2.14)	-0.02 (-1.88)	-0.15 (-0.36)	-0.14 (-0.33)	-0.16 (-0.36)
Redeployability (aircraft)*1000	0.08 (1.90)			0.09 (2.72)			-1.04 (-1.32)		
Redeployability (operators)		0.001 (1.92)			0.001 (2.65)			-0.01 (-1.39)	
Redeployability (>5 aircraft)			0.002 (1.63)			0.002 (2.34)			-0.03 (-1.44)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Airline fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.90	0.90	0.90	0.93	0.93	0.93	0.56	0.56	0.56
Observations	207	207	207	76	76	76	230	230	230
Unit of observation	tranche	tranche	tranche	issue	issue	issue	tranche	tranche	tranche

Table 10: Collateral value, loan-to-value, and tranche maturity.

Table 11
Economic significance of collateral redeployability.

Predicted changes in the dependent variables as each redeployability measure varies (i) by one standard deviation, (ii) from the 25th percentile to the 75th percentile, (iii) from the 10th percentile to the 90th percentile. Panel A reports level changes basis points (bp) and percentage changes relative to the sample spread mean (in parentheses). Panel B reports marginal effects of changes in credit rating category probability associated with one standard deviation move, and their corresponding percentage change relative to the sample mean (in parentheses) in the redeployability measures. Panel C reports changes in LTV as well as percent changes relative to the sample LTV mean (in parentheses). The results are computed using the specifications in Tables 6, 8 and 9.

	Standard deviation	25th-75th Percentile	10th-90th Percentile
Panel A: Credit spread			
Redeployability (aircraft)	-40.5 bp (-20.4%)	-64.2 bp (-32.3%)	-99.5 bp (-50.0%)
Redeployability (operators)	-35.7 bp (-18.0%)	-52.2 bp (-26.3%)	-85.8 bp (-43.2%)
Redeployability (>5 aircraft)	-38.3 bp (-19.3%)	-58.0 bp (-29.2%)	-93.0 bp (-47.2%)
	$\partial \text{Prob}(\text{rating} = \text{AA}+) / \partial x$	$\partial \text{Prob}(\text{rating} = \text{A}) / \partial x$	$\partial \text{Prob}(\text{rating} = \text{BBB}+) / \partial x$
Panel B: Credit rating			
$x = \text{Redeployability (aircraft)}$	0.02 (25.6%)	-0.01 (-6.9%)	-0.05 (-35.7%)
$x = \text{Redeployability (operators)}$	0.02 (25.6%)	-0.01 (-6.9%)	-0.05 (-35.7%)
$x = \text{Redeployability (>5 aircraft)}$	0.03 (38.4)	-0.01 (-6.9%)	-0.06 (-42.9%)
	Standard deviation	25th-75th Percentile	10th-90th Percentile
Panel C: Loan-to-value			
Redeployability (aircraft)	0.071 (11.6%)	0.113 (18.4%)	0.176 (28.5%)
Redeployability (operators)	0.091 (14.7%)	0.133 (21.5%)	0.218 (35.3%)
Redeployability (>5 aircraft)	0.060 (9.7%)	0.091 (14.7%)	0.147 (23.9%)

Table 11: Economic significance of collateral redeployability.

6.13 Table 12: Robustness test controlling for loan-to-value

Read:

- Table 12 adds cumulative loan-to-value to the spread regressions.
- LTV is positively related to spreads, particularly with airline fixed effects.
- Redeployability remains negatively related to spreads after controlling for LTV.
- In the airline fixed-effect specifications, the redeployability coefficients are larger in magnitude than in the baseline spread regressions.

Economic message: Redeployability does not matter only because it allows higher LTV. Even conditional on LTV, more redeployable collateral lowers spreads.

6.14 Table A1: Redeployability and airline characteristics

Read:

- Table A1 tests whether airline risk predicts collateral redeployability.
- The dependent variables are the three redeployability measures.
- Explanatory variables include size, profitability, leverage, interest coverage, and airline credit rating.
- The variables are not statistically or jointly significant in explaining redeployability.
- This weakens the concern that riskier airlines systematically pledge more redeployable collateral.

Economic message: The appendix table supports the paper's identification logic. The collateral-quality variation does not appear to be a simple reflection of borrower risk.

Table 12

Robustness test: controlling for loan-to-value.

The dependent variable in the regressions is credit spread (in basis points) over its corresponding treasury yield. Loan-to-value is the cumulative loan-to-value (LTV) at the tranche level. Seniority is the tranche seniority (1 = most senior). Private is a dummy variable that equals one for private placement tranches. Tranche size is the logarithm of the dollar value (in \$ millions) of the tranche. Call provision is a dummy variable that equals one if the tranche is callable. Maturity is the number of years until the final payment. Airline size is the logarithm of the book value of the airline assets. Market-to-book is calculated as the market value of equity minus the book value of equity, all over the book value of assets. Profitability is defined as operating income over assets. S&P airline credit rating is the airline long-term credit rating. Redeployability (aircraft) is the number of aircraft per type; Redeployability (operators) is the number of operators per type; Redeployability (>5 aircraft) is the number of operators who operate at least five aircraft per type. All regressions include an intercept (not reported) and both year and airline fixed effects. *t*-statistics are calculated using standard errors that are clustered by airline and reported in parentheses.

Dependent variable =	Credit spread	Credit spread	Credit spread	Credit spread	Credit spread	Credit spread
Loan-to-value	108.21 (0.72)	110.14 (0.74)	108.98 (0.74)	349.12 (2.00)	364.96 (2.03)	343.61 (2.00)
Seniority	45.02 (7.28)	44.93 (7.32)	44.87 (7.38)	18.04 (1.62)	16.36 (1.39)	18.36 (1.63)
Private	38.23 (3.23)	36.12 (3.16)	36.25 (3.15)	36.25 (2.99)	24.18 (1.72)	29.75 (2.54)
Tranche size	0.40 (0.05)	0.99 (0.12)	0.86 (0.10)	2.94 (0.29)	3.92 (0.45)	2.80 (0.34)
Call provision	66.00 (3.25)	64.96 (3.33)	65.20 (3.07)	55.81 (4.34)	55.69 (4.15)	58.78 (3.53)
Maturity	0.16 (0.09)	0.11 (0.06)	0.08 (0.04)	0.26 (0.13)	0.09 (0.04)	0.16 (0.08)
Airline size	-15.67 (-1.60)	-16.48 (-1.64)	-16.56 (-1.68)	179.35 (2.72)	174.68 (2.31)	152.11 (2.29)
Market-to-book	17.53 (0.72)	19.60 (0.78)	19.23 (0.81)	94.32 (2.90)	107.13 (2.91)	84.20 (2.99)
Profitability	-210.25 (-2.11)	-213.64 (-2.09)	-210.24 (-2.05)	-342.78 (-1.58)	-332.10 (-1.55)	-418.18 (-1.84)
S&P airline credit rating	7.32 (1.76)	7.31 (1.76)	7.19 (1.76)	2.23 (0.51)	2.14 (0.40)	1.37 (0.35)
Redeployability (aircraft)	-0.016 (-2.98)			-0.10 (-4.49)		
Redeployability (operators)		-0.145 (-3.15)			-0.83 (-3.07)	
Redeployability (>5 aircraft)			-0.457 (-3.15)			-2.34 (-4.03)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Airline fixed effects	No	No	No	Yes	Yes	Yes
Adjusted R ²	0.63	0.63	0.63	0.68	0.68	0.67
Observations	203	203	203	203	203	203

Table 12: Robustness test controlling for loan-to-value.

Table A1

Redeployability and airline characteristics.

The dependent variable in the regressions is one of three redeployability measures: Redeployability (aircraft) is the number of aircraft per type; Redeployability (operators) is the number of operators per type; Redeployability (>5 aircraft) is the number of operators who operate at least five aircraft per type. Airline size is the logarithm of the book value of airline assets. Profitability is defined as operating income over assets. Leverage is defined as total debt divided by total assets. Interest coverage is defined as operating earnings before depreciation divided by interest expense. S&P airline credit rating is the airline long-term credit rating. All regressions include an intercept (not reported) and year fixed effects. Also reported are the *p*-values of *F*-tests for the joint significance of the explanatory variables (excluding year fixed effects). *t*-statistics are calculated using standard errors that are clustered by airline and reported in parentheses.

Panel A: Dependent variable: Redeployability (# of aircraft)						
Airline size	-212.00 (-0.70)					-197.68 (-0.92)
Profitability		3780.45 (1.31)				1661.08 (0.65)
Leverage			558.35 (0.31)			1589.76 (0.94)
Interest coverage				386.33 (1.03)		208.80 (0.76)
S&P airline credit rating					-107.11 (-1.05)	-107.56 (-1.20)
F-test (<i>p</i> -value)						0.407
Adjusted R ²	0.21	0.26	0.18	0.23	0.23	0.340
Observations	225	225	223	225	225	225
Panel B: Dependent variable: Redeployability (# of operators)						
Airline size	-24.90 (-0.74)					-22.12 (-0.93)
Profitability		415.07 (1.22)				148.13 (0.51)
Leverage			32.65 (0.16)			173.69 (0.89)
Interest coverage				53.84 (1.28)		38.07 (1.19)
S&P airline credit rating					-12.58 (-1.08)	-10.61 (-0.98)
F-test (<i>p</i> -value)						0.339
Adjusted R ²	0.21	0.22	0.19	0.21	0.24	0.29
Observations	225	245	225	225	225	225
Panel C: Dependent variable: Redeployability (# of operators with >5 aircraft)						
Airline size	-8.54 (-0.76)					-7.95 (-0.99)
Profitability		143.02 (1.31)				58.59 (0.64)
Leverage			13.91 (0.21)			55.04 (0.91)
Interest coverage				15.97 (1.17)		0.06 (0.01)
S&P airline credit rating					-4.12 (-1.08)	-3.95 (-1.31)
F-test (<i>p</i> -value)						0.377
Adjusted R ²	0.23	0.26	0.16	0.25	0.25	0.36
Observations	225	225	225	225	225	225

Table A1: Redeployability and airline characteristics.

7 Short summary

- The paper studies how collateral quality affects secured debt pricing.
- The empirical setting is U.S. airline secured debt issued from 1990 to 2005.
- The key innovation is to measure collateral redeployability using the worldwide secondary market for specific aircraft types.
- The paper studies the intensive margin of collateral use, avoiding much of the selection bias in secured-versus-unsecured comparisons.
- More redeployable collateral is associated with lower credit spreads.
- More redeployable collateral is associated with higher credit ratings.
- More redeployable collateral is associated with higher loan-to-value ratios.
- The paper does not find a robust maturity effect.
- The effects are economically large: 25th–75th percentile differences in redeployability reduce spreads by roughly 52–64 basis points and increase LTV by about 9–13 percentage points.
- The findings support theories in which collateral liquidation value lowers creditors' downside risk and increases debt capacity.

8 Conclusion

8.1 Core conclusion

Benmelech and Bergman show that the value of collateral matters for the pricing and structure of secured debt. Aircraft-backed tranches secured by more redeployable aircraft have lower spreads, better ratings, and higher loan-to-value ratios. This supports the idea that creditors price expected liquidation value, not merely the legal presence of collateral.

8.2 Economic intuition

The central intuition is straightforward. If an airline defaults, creditors care about what they can recover from the pledged aircraft. A Boeing model operated by many solvent airlines has a deeper secondary market than a niche aircraft operated by few buyers. Deeper secondary markets make liquidation easier and reduce expected loss given default. Creditors therefore accept lower spreads and allow higher borrowing against collateral.

8.3 Incremental contribution revisited

The paper advances the collateral literature by moving from a binary measure of collateral to a granular measure of collateral redeployability. It constructs a novel asset-level data set linking debt tranches to the exact aircraft pledged as collateral. This allows the authors to test theories of collateral, liquidation value, and debt capacity using variation in the quality of collateral rather than variation in the existence of collateral.

8.4 Implications for corporate finance

The results imply that asset specificity and secondary-market depth are central to debt capacity. Firms whose assets can be easily redeployed should be able to borrow more cheaply and at higher LTV ratios. Firms with specialized assets may face higher financing costs, even if those assets are legally pledgeable.

8.5 Implications for lenders and rating agencies

The findings suggest that lenders and rating agencies should evaluate collateral using market-depth measures, not only appraised asset values. A highly appraised but specialized asset may be less valuable to creditors than a more standardized asset with many potential users.

8.6 Limitations

The paper studies one industry, so external validity to other asset classes depends on whether similar secondary-market measures can be constructed. The analysis is not randomized; fixed effects and selection tests reduce but cannot eliminate all concerns. Tranche characteristics such as LTV, maturity, and seniority are jointly chosen, which complicates causal interpretation of individual contract terms. The study also measures collateral value at issuance, so it may understate collateral's value over the life of the debt if default risk increases later.

8.7 Takeaway

The key takeaway is that collateral is valuable because it can be redeployed. In secured debt markets, creditors price not only whether they have a claim on assets, but also whether those assets can be sold to many alternative users if the borrower fails.